

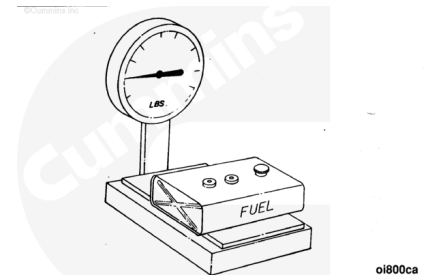
Trainings ▾

Measure

Refer to the fuel consumption checklist sheets in Procedure t00-004 (/qs3/pubsys2/xml/en/procedures/100/100-t00-004.html), in the back of Section TS.

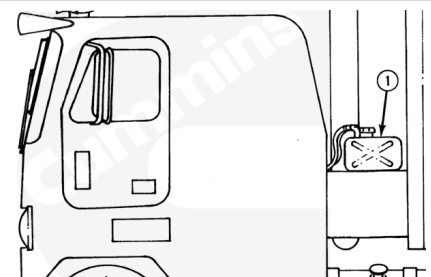
Note : The most accurate method to check the fuel consumption is to weigh the fuel used. Use a scale capable of measuring within 0.045 kg [0.1 lb] to weigh the fuel tank. Use a remote-mount tank with enough capacity to run 80 km [50 mi].

Fill the fuel tank. Weigh the tank with the fuel. The weight on Number 2 diesel fuel is nominally 0.844 kg per liter [7.03 pounds per gallon].

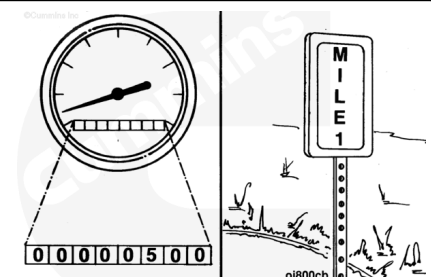


Install the remote tank (1).

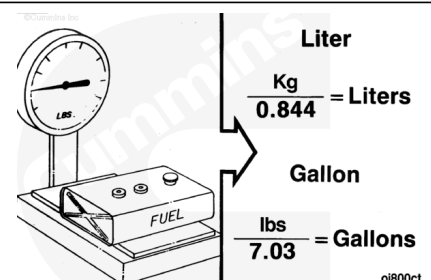
Install the return fuel line to the test tank, or the results will **not** be accurate.



Measure the distance traveled with an accurate odometer. The odometer accuracy can be checked by using measured kilometers [miles].



After traveling the route, remove the tanks, and weigh the remaining fuel. Compute the fuel used in liters [gallons] as required.



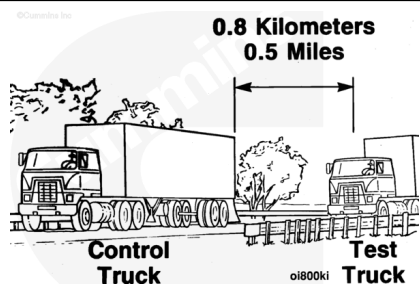
Compute the kilometers per liter or miles per gallon.

$$\frac{\text{Miles}}{\text{Gallons}} = \text{MPG}$$
$$\frac{\text{Kilometers}}{\text{Liters}} = \text{KPL}$$

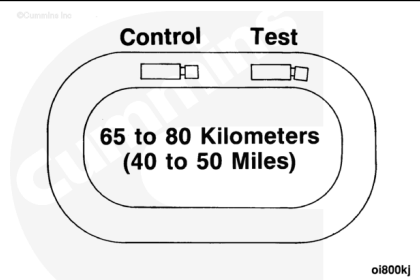
In addition to the measurement of the fuel used, the following factors provide points for running a test similar to the recognized Type II Society of Automotive Engineers Fuel Test. These procedures are helpful to determine differences in fuel consumption between two vehicles under the same environmental, road, and test conditions.

Type II
Fuel Test
SAE
RCCC
ATA

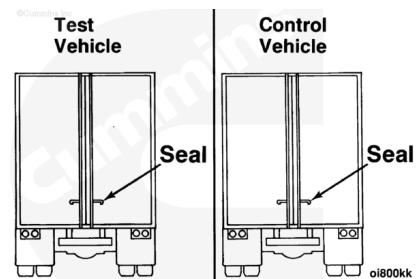
Perform the test with the test vehicle and a control vehicle. The control vehicle compensates for changes in traffic conditions. The vehicles **must** stay close together to experience the same varying traffic and weather conditions, but **not** so close as to affect each other's driving or headwind.



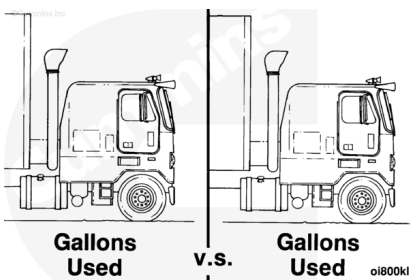
The test course **must** be 65 to 80 km [40 to 50 mi] long.



The test route and truck weights **must not** change during the test.

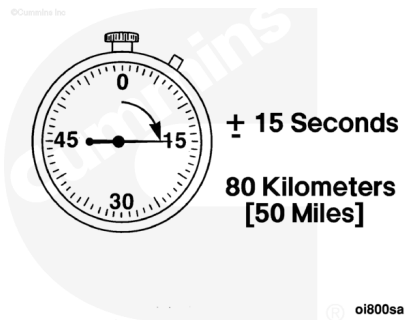


All of the test results are based on comparing the fuel used by the test truck to the fuel used by the control truck.



Drive the truck on a warm-up test run. Drive enough tests to achieve:

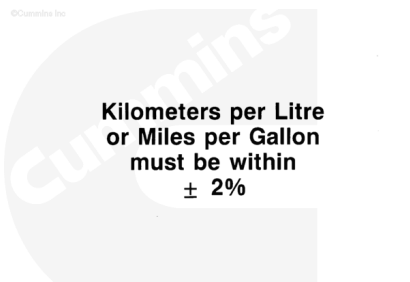
Difference in elapsed time between each test run can only be plus or minus 0.5 percent. This will be ± 15 seconds on 80 km [50 mi] at 60 miles per hour.



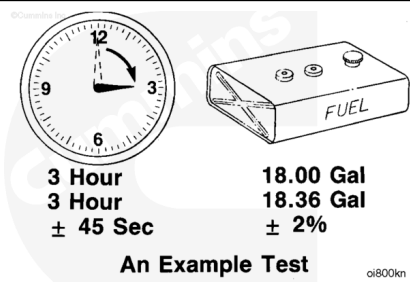
The fuel usage of the test truck between test drives **must** fall within a 2-percent range, e.g., 2.55 vs. 2.60 km/l [6.00 vs. 6.12 mpg].

The same range also applies between drives of the control truck.

Note : The differences in traffic and driving practices can make the test drive fall out of the 2-percent range.

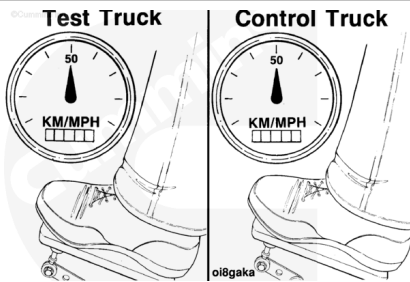


A minimum of three test drives that meet these conditions make a valid test. A single test drive is unreliable.



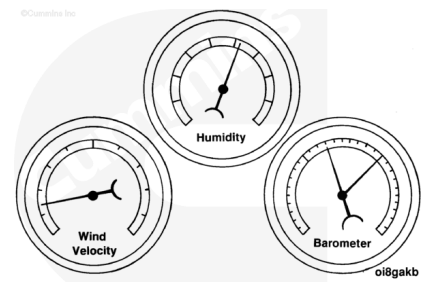
Use the same experienced drivers for all of the tests.

Note : The vehicle speeds must be representative of a typical operation.



During the test, record the following:

- Ambient temperature
- Humidity
- Barometric pressure
- Wind velocity
- Wind direction.



Note : Avoid testing under any extreme conditions.